Contributor Screenshots and Tests

All tests under Javadoc HTML:
/JavaDoc/FoundationsF26/automationTestbed/ContributorTestingAutomation.html

Requirement	Cases	Javadoc method
Add lessons with core information	1–6	testCreateLessons()
Examine a lesson	7–8	testReadLessons()
List and select lessons and support subsets	9–10	testReadLessons()
Manage editable lesson information	11–14	testUpdateLessons()
Delete lessons	15–16	testDeleteLessons()
List only a specified author's lessons	17–18	testAuthorLessonList()
Persistent lesson storage	19	testPersistence()

Create lessons, Tests 1-6

Test method: testCreateLessons()

Purpose: These tests check the user story that a Contributor can add a lesson with core information. Test 1 checks that the title and description are actually saved to the database with the author and a unique ID. Tests 2 and 3 check that required core information cannot be left empty. Test 4 tests unique title rules we implemented, and makes sure a rejected duplicate does not replace the original lesson. Tests 5 and 6 checks the 1600 character description limit, the full allowed amount must save to the database(5), and an extra character must be rejected(6). And a negative case passes when the input is rejected and nothing is saved or changed.

Test 1: Valid lesson is saved correctly

Positive Test

Creates a lesson and reads it back. Passes when the title, description, author, and generated ID are correct.

Screenshot file: testCreateLessons1-4.png

Test 2: Missing description is rejected

Negative Test

Attempts creation with an empty description. Passes when creation returns false and no lesson is saved.

Screenshot file: testCreateLessons1-4.png

Test 3: Missing title is rejected

Negative Test

Attempts creation with an empty title. Passes when creation returns false and no lesson is saved.

Screenshot file: testCreateLessons1-4.png

Test 4: Duplicate title preserves the original lesson

Negative Test

Attempts to reuse an existing title. Passes when the duplicate is rejected and the original values remain unchanged.

Screenshot file: testCreateLessons1-4.png

Test 5: Maximum valid description saves completely

Positive Test

Saves a 1600 character description. Passes when all 1600 characters are retrieved unchanged.

Screenshot file: testCreateLessons5-6.png

Test 6: Description above 1600 characters is rejected

Negative Test

Attempts to save 1601 characters. Passes when creation returns false and no lesson is saved.

Screenshot file: testCreateLessons5-6.png

Read Lessons, Tests 7-10

Test method: testReadLessons()

Purpose: These tests check the lesson data needed for the story that a Contributor can list lessons and choose one to examine. Test 7 checks that selecting an existing lesson by ID returns the expected saved information from the database. Test 8 checks that a missing ID does not return any

lesson. Test 9 checks that the loaded list and a matching subset contain the intended lesson, and test 10 checks that an unmatched search gives an empty subset. Passing means the returned information or absence matches the expected result.

Test 7: Saved lesson details can be retrieved

Positive Test

Retrieves the saved lesson by ID. Passes when all stored fields match the expected lesson.

Screenshot file: testReadLessons7-10.png

Test 8: Nonexistent lesson returns no details

Negative Test

Requests nonexistent ID -1. Passes when no lesson is returned.

Screenshot file: testReadLessons7-10.png

Test 9: Loaded list and matching subset contain the saved lesson

Positive Test

Loads the list and searches for the test lesson. Passes when the intended lesson is present and the matching subset contains exactly that lesson.

Screenshot file: testReadLessons7-10.png

Test 10: Unmatched search gives an empty subset

Negative Test

Searches for a term with no match. Passes when the result is empty.

Screenshot file: testReadLessons7-10.png

Manage Update Lesson Information, Tests 11-14

Test method: testUpdateLessons()

Purpose: These tests check the story that a Contributor can manage a lesson after adding it. Test 11 checks that changing the title only keeps the same lesson ID, description, and author. Test 12 checks that changing the description only keeps the other information. Tests 13 and 14 check that trying to clear either required field does not damage the existing lesson. Reading the lesson back confirms that valid edits were stored and invalid edits left all fields unchanged.

Test 11: Valid title update is saved

Positive Test

Updates the title and reads the lesson back. Passes when the new title is saved and the ID, description, and author are preserved.

Screenshot file: testUpdateLessons11-14.png

Test 12: Valid description update is saved

Positive Test

Updates the description and reads the lesson back. Passes when the new description is saved and the ID, title, and author are preserved.

Screenshot file: testUpdateLessons11-14.png

Test 13: Empty title update preserves the saved lesson

Negative Test

Attempts an empty title update. Passes when the update is rejected and all stored fields remain unchanged.

Screenshot file: testUpdateLessons11-14.png

Test 14: Empty description update preserves the saved lesson

Negative Test

Attempts an empty description update. Passes when the update is rejected and all stored fields remain unchanged.

Screenshot file: testUpdateLessons11-14.png

Delete a lesson, Tests 15-16

Test method: testDeleteLessons()

Purpose: These tests check deletion as part of managing a lesson. Test 15 checks that the deleted lesson cannot still be retrieved from the database or found in the refreshed list. Test 16 checks that

trying to delete it again is rejected instead of reporting another successful deletion. Passing requires the correct return value and the lesson to remain absent.

Test 15: Deletion removes the lesson from storage and the list

Positive Test

Deletes the prepared lesson by ID and refreshes the list. Passes when deletion returns true and the lesson is absent from both.

Screenshot file: testDeleteLessons15-16.png

Test 16: Already deleted lesson cannot be deleted again

Negative Test

Repeats deletion of the same lesson ID. Passes when deletion returns false and the record remains absent.

Screenshot file: testDeleteLessons15-16.png

List only the author's lessons, Tests 17-18

Test method: testAuthorLessonList()

Purpose: These tests support the stories that a Contributor can list their own lessons and cannot see submissions from other Contributors. Test 17 creates lessons for two authors, then checks that the requested author's list contains both of their lessons and none from the other author. The titles are in opposite alphabetical order so the test also checks our database's ID ordering. Test 18 checks that an author with no lessons receives an empty list instead of someone else's lessons or an error. Passing requires the exact expected list.

Test 17: Owned lessons appear in ID order without another author's lesson

Positive Test

Creates lessons for two authors. Passes when only the requested author's two titles are returned, ordered by ID.

Screenshot file: testAuthorLessonList17-18.png

Test 18: Author with no lessons receives an empty list

Negative Test

Requests lessons for an author with none. Passes when a non-null empty list is returned.

Screenshot file: testAuthorLessonList17-18.png

Lesson information persistence between executions, Test 19

Test method: testPersistence()

Purpose: This test checks the database's persistent storage so a lesson added by a Contributor remains available after the program closes. The first run creates a dedicated lesson and leaves it saved within the database. Test 19 on a later run checks if the lesson by title exists, then checks its title, description, author, and positive ID for any edit or missing values since the last run. Passing means those saved values are still available from a separate execution. Creating the record on the first run only prepares this check and is not counted as proof of persistence.

Test 19: Lesson persists between executions

Positive Test

Checks for the persistence lesson saved by a previous execution. If it does not exist, attempts to create it and asks the user to run the tests again for the persistence check.

Screenshot file: testPersistence19.png